

# Yizheng Xie

yizheng\_xie@brown.edu | 857-498-3205 | yizhengx.github.io | Providence, RI

## Education

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| <b>Ph.D. in Computer Science</b> , Distributed Systems, Operating Systems, Cloud Computing<br><b>Brown University</b> , Providence, RI, USA (Advisor: Nikos Vasilakis)  | Sept 2023 – Present  |
| <b>M.S. in Computer Science</b> , Database Systems, Streaming Systems<br><b>Boston University</b> , Boston, MA, USA (Advisors: Vasiliki Kalavri, John Liagouris)        | Sept 2021 – May 2023 |
| <b>B.E. in Electronic Information Engineering</b> , Computer Systems, Machine Learning<br><b>Chinese University of Hong Kong, Shenzhen</b> , Shenzhen, Guangdong, China | Sept 2017 – May 2021 |

## Working Experience

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| <b>Google</b> , Sunnyvale, CA<br><i>Software Engineer Intern in Machine Learning Cluster Runtime, AI Engines Team</i>                                                                                                                                                                                                                                                                                                                                       | May 2026 – Aug 2026 |
| <ul style="list-style-type: none"><li>Improved MLCR's reliability and integration with Pathways for in-memory suspend and resume of TPU workloads.</li><li>Designed and built a TPU efficiency simulator that models context switching across TPU workloads to analyze and quantify TPU efficiency improvements and progress efficiency trade-offs.</li></ul>                                                                                               |                     |
| <b>Meta Platforms</b> , Menlo Park, CA<br><i>Production Engineer Intern in Kernel Team</i>                                                                                                                                                                                                                                                                                                                                                                  | May 2022 – Aug 2022 |
| <ul style="list-style-type: none"><li>Improved and integrated Fedora and CentOS Stream forges, issue trackers, and build workflow into Meta internal tools.</li><li>Implemented a CLI tool to automate the recursive updating of Rust RPM packages, optimizing parallel chain-build order and ensuring full compatibility (Pagure Repository: <a href="https://pagure.io/fedora-rust/rust-update-set">pagure.io/fedora-rust/rust-update-set</a>).</li></ul> |                     |

## Publications

\* Equal contribution.

- [1] **Yizheng Xie\***, Evangelos Lamprou\*, Jerry Xia\*, and Nikos Vasilakis. *Incr: Faster Re-execution via Bolt-on Incrementalization*. In 20th USENIX Symposium on Operating Systems Design and Implementation (**OSDI**), Seattle, WA, July 2026. USENIX Association. ([Paper](#), [Code](#))
- [2] **Yizheng Xie**, Di Jin, Oğuzhan Çölkesen, Vasiliki Kalavri, John Liagouris, and Nikos Vasilakis. *Slowpoke: End-to-end Throughput Optimization Modeling for Microservice Applications*. In 23rd USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Renton, WA, May 2026. USENIX Association. ([Paper](#), [Code](#))
- [3] Zhicheng Huang\*, Ramiz Dundar\*, **Yizheng Xie**, Konstantinos Kallas, and Nikos Vasilakis. *Fractal: Fault-Tolerant Shell-Script Distribution*. In 23rd USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Renton, WA, May 2026. USENIX Association. ([Paper](#), [Code](#))
- [4] **Yizheng Xie**, Di Jin, and Nikos Vasilakis. *Towards Hybrid Cooperative-Preemptive Scheduling*. In Proceedings of the 13th Workshop on Programming Languages and Operating Systems (**PLOS**), Seoul, Republic of Korea, October 2025. ACM. ([Paper](#))

## Research

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| <b>Slowpoke: End-to-end Throughput Optimization Modeling for Microservices</b><br><i>To appear at NSDI'26</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sept 2023 – Apr 2025 |
| <ul style="list-style-type: none"><li>Proposed a novel linear programming-based performance model to quantify end-to-end throughput improvements of hypothetical microservice optimizations without requiring white-box knowledge.</li><li>Developed the first system to accurately predict throughput in complex microservices, achieving RMSE of 2.1% across several real-world benchmarks and 1.9% across 108 synthetic benchmarks with diverse characteristics.</li></ul>                                                                                                                             |                      |
| <b>Fractal: Fault-Tolerant Shell-Script Distribution</b><br><i>To appear at NSDI'26</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sept 2023 – Dec 2024 |
| <ul style="list-style-type: none"><li>Developed a system for fault-tolerant distribution of unmodified shell scripts, delivering average speedups of <math>9.6\times</math> over Bash, <math>5.5\times</math> over Hadoop Streaming (AHS), and <math>1.2\times</math> over DiSH, the state-of-the-art shell scaleout system.</li><li>Designed a lightweight fault-recovery mechanism—including remote-pipe instrumentation and dynamic result persistence—that achieves recovery within <math>1.26\times</math> of fault-free execution and delivers a <math>9.3\times</math> speedup over AHS.</li></ul> |                      |
| <b>Incr: Faster Incremental Execution, Opaquely</b><br><i>To appear at OSDI'26</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Aug 2025 – Jan 2026  |

- Developing a runtime system that automatically memoizes transient results and traces system calls to capture fine-grained and side-effectful dependencies, enabling orders-of-magnitude faster incremental development while providing correctness guarantees.

#### **leash: Scaling Out Shell Scripts with Resumable Cloud Computing**

Sept 2023 – Present

*In preparation (NSDI'27), with Nikos Pagonas, Haoran Zhang and Konstantinos Kallas*

- Developed a system to scale out unmodified shell scripts using cloud (serverful and serverless) computing, achieving  $12.7\times$  average and  $154.2\times$  peak speedups over Bash; in some cases both faster and cheaper, *e.g.*, achieving a  $6.5\times$  speedup while reducing cost by 16.4% compared to PASH—the state-of-the-art shell parallelization system.
- Designed a *suspend-and-resume* protocol that extends AWS Lambda beyond the 15-minute execution limit while preserving correctness and Unix-style streaming efficiency, with negligible slowdown (avg: 3%).

#### **MeshBench: Benchmarking and Analyzing Service Meshes for Microservices**

Aug 2025 – Present

*Under Submission (NINEs'27), with Alice Song, Deepti Raghavan and Akshay Narayan*

- Developing a comprehensive benchmarking suite to evaluate the performance overhead and trade-offs of leading service meshes—*e.g.*, Istio, Cilium, Ambient, Canal—in microservice environments, and analyzing how service mesh architectures and functionalities, *e.g.*, security and observability, affect performance and resource utilization.

### **Invited Talk**

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#### **Towards Automated Performance Optimizations for Service-Oriented Architectures**

Oct 2025

*Boston University, Boston, MA, USA*

### **Teaching**

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#### **CSCI 1380 Distributed Systems**, Brown University, *Teaching Assistant (Spring 2023, Spring 2024)*

- Co-designed and implemented hands-on assignments for building a large-scale search engine that emphasized core distributed systems concepts, including a distributed key-value store for scalable data storage, data sharding for load balancing, consensus protocols for data consistency, and a MapReduce system for efficient data processing.

#### **CSCI 2952R Systems Transforming Systems Seminar**, Brown University, *Teaching Assistant (Fall 2024)*

- Mentored students in conducting state-of-the-art systems research across operating systems, distributed systems, and cloud computing, guiding them through system implementation, experimental evaluation, and academic paper writing.

### **Skills**

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- **Programming Languages:** Go, Python, C/C++, Rust, Java, JavaScript, Shell, SQL
- **Systems:** Kubernetes, Docker, Istio, AWS (*e.g.*, Lambda), Linux/Unix, Apache Flink, Spark, V8/libUV, MongoDB, Redis
- **Tools:** Git, CI/CD (GitHub Actions, GitLab CI), Helm, Make, CMake, LaTeX, Vim, perf

### **Languages and Interests**

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- **Languages:** Mandarin (native), English (fluent)
- **Interests:** Photography, Video Editing, Badminton, Basketball, Traveling, East Asian History